Keel Classification Scheme

Oct 23, 2026 · @jeffrey

Design principles

Classify meaning once, on the concept, and let the mapping graph carry that classification to every place the concept appears. A column, a document, a vector chunk, a metric and an AI answer all inherit from the concepts they map to, so nobody labels the same fact twice.

1. **Classify concepts, not strings.** `pes:Worker` date of birth is Restricted wherever it lives; a regex for "DOB" is only a fallback for text that cannot be anchored to a URI.
2. **Facets, not one ladder.** Sensitivity, data subject, tenancy, regime, purpose, time and provenance vary independently and are recorded separately.
3. **High-water mark.** Anything built from several inputs takes the highest sensitivity among them. Automatic reclassification may only raise a level; lowering one needs a steward and a KGCL change record.
4. **Unknown means held.** An asset or term that does not resolve through the mapping graph is treated as Restricted until a steward classifies it.
5. **Define in the graph, enforce at runtime.** Classifications are authored as RDF and SHACL, then compiled into Purview labels, OPA attributes, vector filters and column masks, per the Compiled Semantic Governance pattern.

Facets

Seven facets make up a classification; together they answer the 5D question of who may see what, where, when and why. Each facet is its own SKOS concept scheme in `isp-class`.

Facet	5D dimension	Values (examples)	Combines by
Sensitivity	WHAT	C0 Public, C1 Internal, C2 Confidential, C3 Restricted, C4 Highly restricted	Highest wins

Facet	5D dimension	Values (examples)	Combines by
Data subject	WHO (about)	Worker, client employer, Insperity, third party, none (aggregate)	Union
Tenancy	WHERE	Tenant-scoped (one client), cross-tenant aggregate, Insperity-owned	Narrowest scope wins
Regime	WHY (legal basis)	HIPAA plan PHI, ADA/FMLA medical confidentiality, BIPA biometric, CCPA/CPRA, FCRA background check, Form I-9, attorney-client privilege, MNPI	Union; every regime's rules apply
Permitted purpose	WHY (use)	Payroll and tax, benefits administration, compliance, advisory, analytics, AI grounding	Intersection
Temporal	WHEN	Effective from/to, embargo until, retention end, legal hold	Latest start, earliest end
Provenance	WHO (made it)	System of record, human-authored, AI-generated, AI-derived aggregate	Least trusted wins

The "Combines by" column is what lets the gateway compute a classification for anything assembled from parts, including AI answers.

Sensitivity levels

Five ordered levels; C3 and C4 cover most worker data, so most of the gateway's work happens there.

Level	Meaning	PEO examples	Default handling
C0 Public	Published or safe to publish	Statute text, published minimum wage rates, public plan summaries	Any audience

Level	Meaning	PEO examples	Default handling
C1 Internal	Insperity business use	Service procedures, rule register, ontology itself	Insperity staff; client-facing only as summaries
C2 Confidential	A client's or Insperity's business information	Client service agreement terms, client headcount, pricing, aggregate metrics with n ⩾ 5	Same tenant, named roles
C3 Restricted	Identifies or describes a worker	Name with pay rate, date of birth, home address, performance ratings, leave balances	The worker, plus roles with a stated purpose
C4 Highly restricted	Harm if exposed; special legal regime	SSN and government IDs, bank accounts, health plan elections and claims, medical leave reasons, biometrics, background checks, investigations, privileged legal advice, MNPI	Named role and purpose only; never in AI grounding without redaction

MNPI (material non-public information) sits at C4 because Insperity is publicly traded: unreleased book-of-business metrics must not surface through a client-facing advisor.

Where classification attaches

Classification is set at four levels, and the effective value is the highest one that applies. Most of it is authored once at the concept level; the other three handle exceptions and combinations.

Level	Set on	Who sets it	Example
Concept	An OWL class or property in an L1/L2 holon (`isp-class:defaultSensitivity`)	Domain Governance Team, via ORSD	`isp-pay:payRate` → C3; `isp-ben:medicalElection` → C4, HIPAA

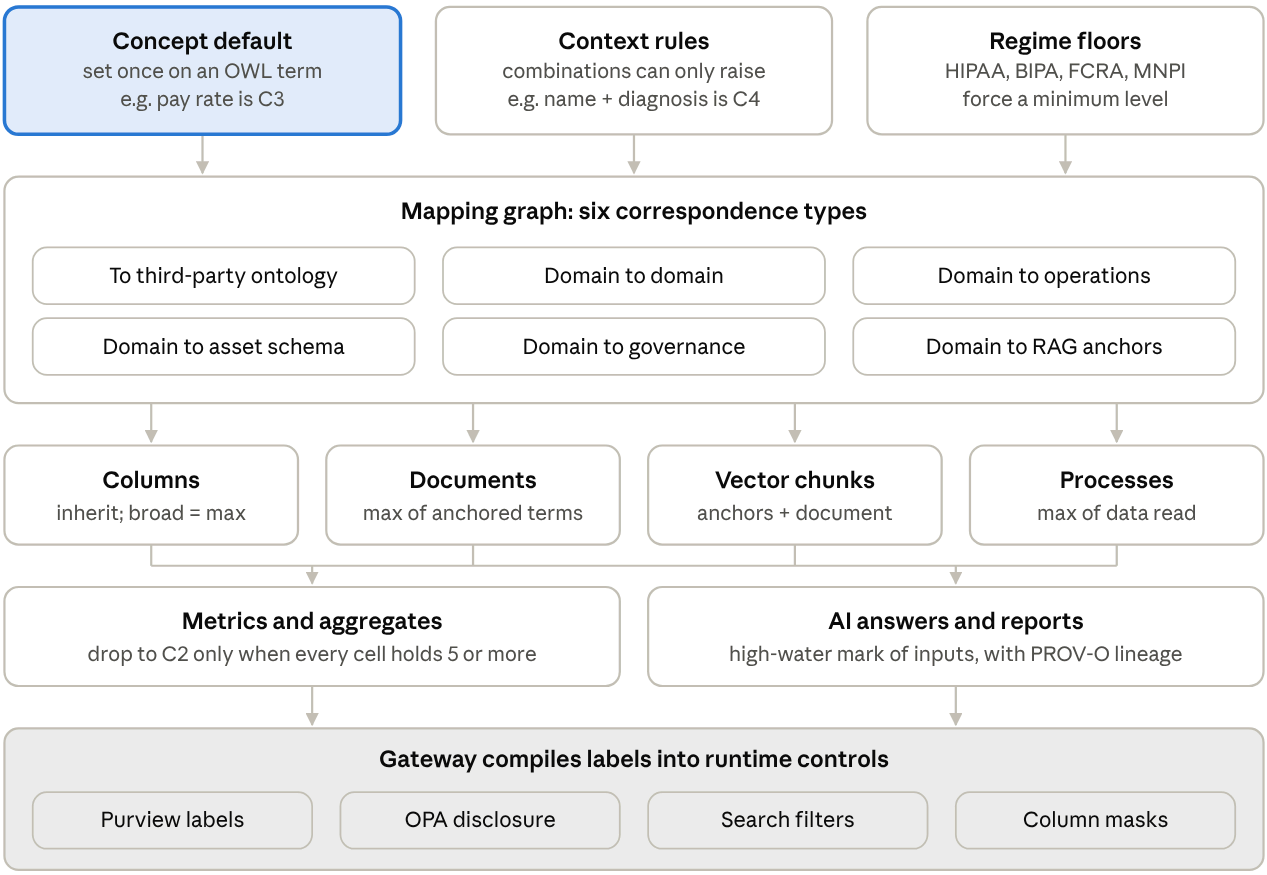
Level	Set on	Who sets it	Example
Instance	One record, document or file, raise only	Steward, or an admission rule	A handbook draft under legal review → C4, privileged
Context	A combination of concepts (a SHACL rule)	Semantic Governance Council	Worker name alone → C3; name + diagnosis → C4; job title + pay in a team of one → C3
Derived	Anything computed or generated from inputs	The gateway, at run time	An AI answer citing a C3 fact → C3

The context level handles the mosaic effect: individually harmless facts that identify a person when combined. These rules are written as SHACL-SPARQL constraints so the Rule Harmonization Engine can check them for conflicts.

Propagation through the mapping topology

The mapping graph is the propagation path: every mapped item inherits from the terms it maps to, and anything that resolves through no validated mapping is held at C3.

Classify the concept once; the mapping graph carries it everywhere



Classification propagation · concept to mapped items to runtime

Classification flows down only; nothing below the concept layer can lower a level on its own.

Correspondence type	Propagation rule
Ontology to third-party ontology	External terms (ISO 30400, carrier codes) never set Insperity sensitivity; the Insperity term classifies the data
Domain to domain	Exact or close matches must carry the same default; a mismatch goes to the Council as a harmonization conflict
Domain to operations	A process or service takes the highest level of the data it reads, which sets who may run it

Correspondence type	Propagation rule
Domain to asset schema	Exact or close: inherit. Narrower asset: at least the term′s level. Broader asset, such as a free-text notes field: the highest default among the term′s narrower concepts
Domain to governance	Links a term to the SHACL shape, LKIF norm and regime that classify it; the audit URI for every compiled policy
Domain to RAG anchors	A chunk takes the highest level of every term anchored in it and of its source document

Only validated mappings may lower an item′s level. A proposed mapping from a Lineage Ingestion Agent can raise one immediately, which keeps new data safe while it waits for review.

Generated content and aggregates

Every AI answer, report and metric is classified before it leaves the gateway, using the same facets as its inputs.

- **Answers and reports.** Sensitivity = the highest of the cited inputs; regimes = their union; provenance = AI-generated. The output carries `prov:wasDerivedFrom` links to each input URI, so a reviewer can see why it got its label.
- **Aggregates.** A count, average or rate drops to C2 only when every cell holds at least the minimum cell size (default 5). Below that, the cell inherits its members′ level (usually C3) and is suppressed or rolled up for audiences without C3 access.
- **Complementary suppression.** When one cell is suppressed, a second is suppressed too, so the hidden value cannot be recovered from the total.
- **Small clients.** For a client under the minimum size, any per-person metric is C3 by definition. The advisor offers benchmarks against peer companies instead.
- **Labels travel.** Generated files carry the matching Purview sensitivity label, and chat answers show the level so users know how to handle them.

Disclosure matrix

The default decision for each level and audience; regime, purpose and time facets can only tighten it. Tenancy is checked first: nothing from another client's tenant reaches any client audience.

Level	Worker (self)	Client manager	Client owner or HR admin	Insperity service staff	Advisor AI grounding
C0 Public	Allow	Allow	Allow	Allow	Allow
C1 Internal	Summary	Summary	Summary	Allow	Allow
C2 Confidential	Deny	Allow, own team	Allow	Allow, assigned clients	Allow, same tenant
C3 Restricted	Own records	Own reports, by purpose	Allow, by purpose	Allow, assigned clients	Redact identifiers unless the asker may see them
C4 Highly restricted	Own records	Deny	Deny, except a named regime role	Named role only	Deny; route to a person

Two co-employment cases drive the C4 row. A client owner employs the worker but should not see medical leave reasons or health plan claims, which ADA, FMLA and HIPAA confine to a separate file. A client manager sees pay for direct reports only, and only for a stated purpose such as a compensation review.

Enforcement

Classifications are compiled into each port's native controls, and every compiled artifact keeps the URI of the concept classification it came from. No engine re-derives a label on its own.

Port	Compiled target (Microsoft adapter)	What it enforces
Policy	OPA/Rego attributes behind API Management	The disclosure matrix, per request, using the 5D bundle
Analytics	Fabric and dbt column masks, row-level security, minimum-cell tests	Pay, IDs and health fields masked; small cells suppressed

Port	Compiled target (Microsoft adapter)	What it enforces
Grounding	Azure AI Search metadata fields and filters	A chunk is retrieved only if the asker's level, tenant and purpose allow it
Content	Purview sensitivity labels and DLP rules on Box and Microsoft 365	Labels follow files out of the platform
Agents	JSON Schema output contract	Every answer must carry its level, regimes and source URIs
Telemetry	PROV-O records, Sentinel alerts	Every release and denial is logged against its classification

Purview stays the outer perimeter. The Keel adds what Purview labels cannot express: co-employment roles, per-client tenancy and context rules.

Worked example

The owner of a 12-person design studio asks the advisor, "What's the average pay of my designers, and is it competitive?" The answer is released, but only in a form that does not expose anyone.

1. **Resolve.** WHO = client owner; WHAT = `isp-pay:payRate` for job family Designer; WHERE = their tenant, worksite in New York City; WHY = advisory; WHEN = current pay period.
2. **Classify inputs.** `isp-pay:payRate` is C3 at the concept level. Three designers are on payroll, below the minimum cell size of 5.
3. **Decide.** The owner may see C3 pay data for their own workers for an advisory purpose, so individual rates are allowed. The average of three is also allowed, since the asker can see each member.
4. **Benchmark.** The comparison uses a peer group of similar New York City companies. That cross-tenant aggregate is C2 only because it meets the minimum cell size and contains no tenant identifiers.
5. **Label the answer.** Output = C3 (highest input), regime = none, provenance = AI-generated, with `prov:wasDerivedFrom` to the payroll records and the benchmark.

If a designer asked the same question, step 3 would deny the team average: a worker may see only their own pay, and three people is too few to aggregate.

Governance, lifecycle and ontology files

The scheme follows the same ORSD → KGCL → SHACL lifecycle as every other holon. Two files carry it today.

File	Contents
`isp-classification.ttl`	Seven facet schemes (SKOS), the classification vocabulary (OWL 2 DL), the six correspondence types with their propagation rules, concept defaults for sample `pes`, `isp-pay` and `isp-ben` terms, and worked-example assets
`isp-classification-shapes.ttl`	Eight SHACL gates: one classification per item; required facets; tenant named; high-water mark; regime floor; concept floor; minimum cell size; unmapped items held

Who does what

- Domain Governance Teams set concept defaults in their holons through the ORSD.
- The Semantic Governance Council approves context rules, regime floors and the minimum cell size.
- Stewards clear the held queue and approve any lowering, each recorded in KGCL.
- Lineage Ingestion Agents propose mappings for new columns and documents; a proposed mapping may raise a level but only a validated one may lower it.

Open items

- [ ] Confirm the Purview label names with Information Security.
- [ ] Confirm the minimum cell size (5) with Legal and Privacy.
- [ ] Move the `isp-pay` and `isp-ben` sample terms into their domain holons once those exist.
- [ ] Run the shapes through a full SHACL engine; they were checked only for syntax here, because package downloads were blocked.